

3) If you make home videos with a camera, use your 328 to create lead in titles, credits, etc... This will give your videos that "professional" flavour.

4) Create a tape catalog at the beginning of your VCR tapes. Leave a minute or so of blank space at the beginning of the tape, then use your word processor, or your own computer program to make the catalog. Include the names of the shows, their length, and counter numbers to indicate where they begin and end. You might even want to include your own mini-review of a movie!

USE OF THE SPECTRAVIDEO RS232 INTERFACE TO CONTROL A SERIAL PRINTER

Reprinted from: SVI NEWS & VIEWS - By: Bernard L. Burke
FROM: South Africa

The SV 805 is a standard RS232 interface which can be used to connect your computer to a SERIAL PRINTER. The SV 805 can also be used to transfer files between computers directly or via the telephone system through a MODEM.

The function of an RS232 interface is to control the DATA FORMAT, and the DATA TRANSMISSION RATE.

With SERIAL DATA TRANSFER the computer sends out batches of BITS down a single data line (NB: with PARALLEL transfer there are 8 data lines one for each BIT of a character). Each batch of bits represents a single character.

A batch of bits consists of the following: 1) A START BIT; 2) 5 to 8 DATA bits; 3) A PARITY bit (optional); 4) 1 or 2 STOP bits.

1) The START bit is automatically generated by the RS232 and so the user need not worry about it.

2) The user must select the required of DATA bits - also known as WORD LENGTH. 7 data bits are sufficient to transmit all the ASCII characters, but if you want to use any special character you will need 8 bits.

3) The parity bit is an optional bit which may be added as a means of error checking. It works like this: a) EVEN parity - the parity bit is set such that there will be an even number of bits (data word + parity) set to "1". The receiving device then checks the number of "1" bits to detect any errors. b) ODD parity - this is similar to even parity except that an ODD number of "1" bits are transmitted and checked. c) NO parity - the parity bit is disabled. d) MARK parity - the parity bit is always set to "1". e) SPACE parity - the parity bit is always reset to "0".

4) The STOP bits tell the receiver that a complete DATA WORD (character) has been received.

The BAUD rate is the rate at which data is transmitted. BAUD rate is measured in BITS PER SECOND (BPS).

The SV 805 has 2 programmable registers for setting up the desired conditions - these are the LINE CONTROL REGISTER and the INTERRUPT ENABLE REGISTER. In addition there is a BAUD RATE GENERATOR which must be programmed.

Proceed as follows:

- 1) Decide on the baud rate required (generally dictated by the receiving device);
- 2) Calculate the value D using the formula: $D = 192000/\text{BAUD RATE}$;
- 3) Divide D into a HIGH BYTE (DH) and a LOW BYTE (DL) where: $DH = \text{INT}(D/256)$ $DL = (DH \times 256)$;

- 4) Decide on the WORD LENGTH required - WORD LENGTH can be 5,6,7 or 8 bits. Calculate WL where: $WL4 = \text{WORD LENGTH} - 5$;
- 5) Select the number of STOP BITS required - one or two STOP BITS are allowed. Calculate SB where: $SB = 4 * (\text{NUMBER OF STOP BITS} - 1)$;
- 6) Decide on which PARITY mode is required - select PB such that:

NO PARITY	PB=0
ODD PARITY	PB=8
EVEN PARITY	PB=24
MARK PARITY	PB=40
SPACE PARITY	PB=56

7) Now calculate $CL = WL + SB + PB$; 8) The Z80A interrupts cannot be controlled from basic so the SV 805 interrupt register should be disabled -IR = 0; 9) Having set all the variables we are now ready to program the SV 805. Enter the following as a program line:
 100 OUT &H2B,&H80 : OUT &H29,DH : OUT &H28,DL : OUT &H2B,CL :
 OUT &H29,IR

When entering the line you should replace the variables DH, DL, CL and IR with their respective values.

When this line is executed your SV 805 is initialised according to the chosen values.

IMPORTANT: THE SENDING DEVICE AND THE RECEIVING DEVICE MUST BOTH BE INITIALISED TO THE SAME CONFIGURATION - FAILURE TO DO THIS WILL CAUSE THE DATA TO BECOME GARBLED DURING TRANSFER.

Data transmission is through OUTPUT PORT &H28 (eg. OUT &H28,X) - all data must be in ASCII code.

IMPORTANT POINTS TO NOTE

a) Ensure that the previous character has been read by the printer before sending the next character - this is done by reading bit 5 of the MODEM STATUS REGISTER. If bit 5 is "0" then wait until bit 5 is "1". The modem status register is read from INPUT PORT &H2E - use the following code to read bit 5:

```
1000 IF (INP(&H2E)AND 32)=0 THEN 1000
```

This program line will loop until the printer is ready.

b) Ensure that the TRANSMITTER HOLDING REGISTER is empty before sending the next character - this is done by reading bit 5 of the LINE STATUS REGISTER. If bit 5 is "0" then wait until bit 5 is "1". The line status register is read from INPUT PORT &H2D - use the following code to read bit 5:

```
1010 IF (INP(&H2D)AND 32)=0 THEN 1010
```

This program line will loop until the transmitter holding register is empty.

LIST 1 is a simple RS232 PRINTER program for your SPECTRAVIDEO. The program illustrates the initialisation of the SV 805 and the transfer of TEXT to the serial printer. The program uses a BAUD RATE of 9600, WORD LENGTH 8 bits, NO PARITY bit, and 2 STOP bits. If your printer requires different values then change the values in LINE 100.

LIST 1

```
10 CLS: INPUT "TYPE IN TEXT TO BE PRINTED";AS
100 OUT &H2B,&H80: OUT &H29,0: OUT &H28,20: OUT &H2B,7: OUT &H29,0
900 AS=AS+CHR$(13)+CHR$(10)
910 FOR Z=1 TO LEN(AS): X=ASC(MID$(AS,Z,1))
1000 IF (INP(&H2E)AND 32)=0 THEN 1000
1010 IF (INP(&H2D)AND 32)=0 THEN 1010
1020 OUT &H28,X:NEXT:GOTO 10
```

RS232 cable configurations differ according to the devices to be connected. An IMPORTANT consideration is that the receiving device must be able to send control signals back to the transmitter. The SV 805 expects such signals on pin 6 but the printer may transmit the signals from pin 20, pin 11 or possibly another pin. Normal data is TRANSMITTED from pin 2 and RECEIVED on pin 3.

The following cable configuration should be suitable for most printers - use SUB-MIN D-TYPE 25-PIN PLUGS.

PIN 7	PRINTER INTERFACE
PIN 20	PIN 7
PIN 6	PIN 6
PIN 4	PIN 11 AND PIN 20
PIN 8	PIN 8
PIN 3	PIN 4
PIN 2	PIN 2
PIN 1	PIN 3
	PIN 1

IN ADDITION YOU SHOULD CONNECT TOGETHER PINS 4 and 5 AT THE SV 805 END and AT THE PRINTER END.

This cable is known as aMODEM ELIMINATORand is only suitable for DIRECT connection of two RS232 devices. When a MODEM is used a different cable configuration will be required.

MSX PRINTERS AND CABLES

REPRINTED FROM: MSX USER GROUP'S NEWSLETTER
FROM: ENGLAND

A reminder that any printer with a parallel interface will work with the MSX micro. Only an MSX printer, however, can print the special MSX graphics characters. These are printed out as spaces on a non-MSX printer. MSX printer cables may not be readily available in shops but can be ordered through a computer dealer.

If you would like to make up your own lead you will need a length of 14 strand ribbon cable, a 14 pin plug for the computer and a standard 32-pin centronics plug for the printer. For connection diagram see diagram on front cover.

ONE LINERS
Reprinted from: RESEAU MSX MAGAZINE
FROM: BELGIUM

```
100 COLORC,15,C:SCREEN2:PI=3.14159:A=RND(-TIME):FORZ=0TO360STEP20:X=128+30*
COS(PI*Z/180):Y=96+30*SIN(PI*Z/180):LINE(X,Y)-(128,191):LINEX+5,Y+5)-(0,96)
:LINE(X,Y)-128,0):LINE(X+5,Y+5)-(255,96):NEXTZ:C=(C+1)MOD14:GOTO100

100 COLOR15,1,1:SCREEN2:FORI=1TO100:A=INT(RND(1)*256):B=INT(RND(1)*192):C=
INT(RND(1)*15)+1:LINE(A,B)-(A+50,B+50),C,BF:NEXTI:FORI=1TO10000:1:NEXT

100 COLOR,15,15:SCREEN2:C=(C+1)MOD14:FORX=191TO0STEP-6:LINE(32,X)-(X+32,19
1),C:LINE(223,X)-(X+32,0),C:NEXT:LINE(32,0)-(223,191),C,B:PAINT(1,1),C:PAI
NT(255,1),C:GOTO100
```